



## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

#### Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Other means of identification : Not applicable.

Recommended use : Laundry product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.02 % - 0.1 %

Company  
Ecolab Gulf FZE  
P.O. Box 17063  
Jebel Ali Free Zone Area, Near Container Terminal 3 - North Zone  
Dubai  
UAE  
00971 4 8014444 Customer Services, 00971 48146800 00971 4  
8014000 Reception phone numbers

Nalco Egypt Trading  
5th Settlement, South 90th St.  
The Address Building No 67th – 1st floor  
New Cairo, Cairo, Egypt11835  
0020 2 25 37 1195,

Ecolab Maroc S.A.R.L.  
Batiment, Lot. Mounir 1, Parc Industriel  
Attawfik  
Route 1029  
Sidi Maarouf, Casablanca, Morocco  
00212 22 58 25 30 - 35,

Emergency telephone number : +32-(0)3-575-5555 Trans-European

Issuing date : 02.12.2019

#### Section: 2. HAZARDS IDENTIFICATION

##### GHS Classification

##### Product AS SOLD

Skin corrosion : Category 1A  
Serious eye damage : Category 1

##### Product AT USE DILUTION

Not a hazardous substance or mixture.

##### GHS Label element

##### Product AS SOLD

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Hazard pictograms



Signal Word : Danger

Hazard Statements : Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**  
Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection.  
**Response:**  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician. Wash contaminated clothing before reuse.  
**Storage:**  
Store locked up.  
**Disposal:**  
Dispose of contents/ container to an approved waste disposal plant.

#### Product AT USE DILUTION

Precautionary Statements : **Prevention:**  
Wash hands thoroughly after handling.  
**Response:**  
Get medical advice/ attention if you feel unwell.  
**Storage:**  
Store in accordance with local regulations.

#### Product AS SOLD

Other hazards : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

#### Product AS SOLD

Pure substance/mixture : Mixture

| Chemical Name     | CAS-No.    | Concentration: (%) |
|-------------------|------------|--------------------|
| formic acid       | 64-18-6    | 10 - 30            |
| citric acid       | 77-92-9    | 1 - 5              |
| ammonium chloride | 12125-02-9 | 1 - 5              |

#### Product AT USE DILUTION

No hazardous ingredients

### Section: 4. FIRST AID MEASURES

#### Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do.

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Continue rinsing. Get medical attention immediately.

|                                                             |                                                                                                                                                                                                        |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In case of skin contact                                     | : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately. |
| If swallowed                                                | : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.                                                            |
| If inhaled                                                  | : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.                                                                                                                 |
| Protection of first-aiders                                  | : If potential for exposure exists refer to Section 8 for specific personal protective equipment.                                                                                                      |
| Notes to physician                                          | : Treat symptomatically.                                                                                                                                                                               |
| Most important symptoms and effects, both acute and delayed | : See Section 11 for more detailed information on health effects and symptoms.                                                                                                                         |

#### Product AT USE DILUTION

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.                           |
| In case of skin contact | : Rinse with plenty of water.                           |
| If swallowed            | : Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled              | : Get medical attention if symptoms occur.              |

### Section: 5. FIREFIGHTING MEASURES

#### Product AS SOLD

|                                               |                                                                                                                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media                  | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                                                       |
| Unsuitable extinguishing media                | : None known.                                                                                                                                                                   |
| Specific hazards during firefighting          | : Not flammable or combustible.                                                                                                                                                 |
| Hazardous combustion products                 | : Decomposition products may include the following materials:<br>Carbon oxides<br>nitrogen oxides (NOx)<br>Sulphur oxides<br>Oxides of phosphorus                               |
| Special protective equipment for firefighters | : Use personal protective equipment.                                                                                                                                            |
| Specific extinguishing methods                | : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

### Section: 6. ACCIDENTAL RELEASE MEASURES

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

#### Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

### Section: 7. HANDLING AND STORAGE

#### Product AS SOLD

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

#### Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

### Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Product AS SOLD

##### Components with workplace control parameters

| Components  | CAS-No. | Form of exposure | Permissible concentration    | Basis  |
|-------------|---------|------------------|------------------------------|--------|
| formic acid | 64-18-6 | TWA OEL-RL       | 5 ppm<br>9 mg/m <sup>3</sup> | ZA OEL |

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

|                   |            |              |                                |         |
|-------------------|------------|--------------|--------------------------------|---------|
| formic acid       | 64-18-6    | TWA          | 5 ppm<br>9.4 mg/m <sup>3</sup> | BH OEL  |
|                   |            | STEL         | 10 ppm<br>19 mg/m <sup>3</sup> | BH OEL  |
| formic acid       | 64-18-6    | TWA (8 Hour) | 5 ppm<br>9 mg/m <sup>3</sup>   | TR OEL  |
| formic acid       | 64-18-6    | TWA          | 5 ppm<br>9.4 mg/m <sup>3</sup> | ARE OEL |
|                   |            | STEL         | 10 ppm<br>19 mg/m <sup>3</sup> | ARE OEL |
| ammonium chloride | 12125-02-9 | TWA (Fumes)  | 10 mg/m <sup>3</sup>           | BH OEL  |
|                   |            | STEL (Fumes) | 20 mg/m <sup>3</sup>           | BH OEL  |
| ammonium chloride | 12125-02-9 | TWA (Fumes)  | 10 mg/m <sup>3</sup>           | ARE OEL |
|                   |            | STEL (Fumes) | 20 mg/m <sup>3</sup>           | ARE OEL |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Wear eye protection/face protection.

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.  
Wash face, hands and any exposed skin thoroughly after handling.  
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

#### Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

### Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

#### Product AS SOLD

#### Product AT USE DILUTION

liquid

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

|                                         |                                                            |                 |
|-----------------------------------------|------------------------------------------------------------|-----------------|
| Colour                                  | : clear, light blue                                        | colourless      |
| Odour                                   | : pungent                                                  | not significant |
| pH                                      | : 0.7 - 1.0, (100 %)                                       | 4.5 - 7.5       |
| Flash point                             | : Not applicable., Does not sustain combustion.            |                 |
| Odour Threshold                         | : no data available                                        |                 |
| Melting point/freezing point            | : no data available                                        |                 |
| Initial boiling point and boiling range | : > 100 °C                                                 |                 |
| Evaporation rate                        | : no data available                                        |                 |
| Flammability (solid, gas)               | : Not applicable.                                          |                 |
| Upper explosion limit                   | : no data available                                        |                 |
| Lower explosion limit                   | : no data available                                        |                 |
| Vapour pressure                         | : no data available                                        |                 |
| Relative vapour density                 | : no data available                                        |                 |
| Relative density                        | : 1.036 - 1.056                                            |                 |
| Water solubility                        | : soluble                                                  |                 |
| Solubility in other solvents            | : no data available                                        |                 |
| Partition coefficient: n-octanol/water  | : no data available                                        |                 |
| Auto-ignition temperature               | : no data available                                        |                 |
| Thermal decomposition                   | : no data available                                        |                 |
| Viscosity, kinematic                    | : no data available                                        |                 |
| Explosive properties                    | : no data available                                        |                 |
| Oxidizing properties                    | : The substance or mixture is not classified as oxidizing. |                 |
| Molecular weight                        | : no data available                                        |                 |
| VOC                                     | : no data available                                        |                 |

#### Section: 10. STABILITY AND REACTIVITY

##### Product AS SOLD

|                                    |                                                                                                                           |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                                                             |
| Chemical stability                 | : Stable under normal conditions.                                                                                         |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                         |
| Conditions to avoid                | : None known.                                                                                                             |
| Incompatible materials             | : Chlorine-based bleaching agents<br>Bases<br>Metals                                                                      |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>nitrogen oxides (NOx)<br>Sulphur oxides |

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Oxides of phosphorus

#### Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

##### Potential Health Effects

###### Product AS SOLD

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Health injuries are not known or expected under normal use.

###### Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

##### Experience with human exposure

###### Product AS SOLD

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

###### Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

##### Toxicity

###### Product AS SOLD

###### Product

Acute oral toxicity : Acute toxicity estimate : > 5,000 mg/kg

Acute inhalation toxicity : no data available

Acute dermal toxicity : no data available

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

|                                   |                     |
|-----------------------------------|---------------------|
| Skin corrosion/irritation         | : no data available |
| Serious eye damage/eye irritation | : no data available |
| Respiratory or skin sensitization | : no data available |
| Carcinogenicity                   | : no data available |
| Reproductive effects              | : no data available |
| Germ cell mutagenicity            | : no data available |
| Teratogenicity                    | : no data available |
| STOT - single exposure            | : no data available |
| STOT - repeated exposure          | : no data available |
| Aspiration toxicity               | : no data available |

### Section: 12. ECOLOGICAL INFORMATION

#### Product AS SOLD Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

#### Product

|                                                     |                     |
|-----------------------------------------------------|---------------------|
| Toxicity to fish                                    | : no data available |
| Toxicity to daphnia and other aquatic invertebrates | : no data available |
| Toxicity to algae                                   | : no data available |

#### Components

|                  |                                           |
|------------------|-------------------------------------------|
| Toxicity to fish | : formic acid<br>96 h LC50: > 100 mg/l    |
|                  | citric acid<br>96 h LC50 Fish: > 100 mg/l |

#### Persistence and degradability

##### Product AS SOLD

Readily biodegradable.

##### Product AT USE DILUTION

Readily biodegradable.

#### Bioaccumulative potential

no data available

#### Mobility in soil

no data available

#### Other adverse effects

no data available

### Section: 13. DISPOSAL CONSIDERATIONS



## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

#### Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

#### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

### Section: 14. TRANSPORT INFORMATION

#### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

#### Land transport (ADR/ADN/RID)

UN number : 3412  
UN proper shipping name : FORMIC ACID  
Transport hazard class(es) : 8  
Packing group : II  
Environmental hazards : No  
Special precautions for user : None

#### Air transport (IATA)

UN number : 3412  
Description of the goods : Formic acid  
Class : 8  
Packing group : II  
Environmentally hazardous : No

#### Sea transport (IMDG/IMO)

UN number : 3412  
Description of the goods : FORMIC ACID  
Class : 8  
Packing group : II  
Marine pollutant : No

### Section: 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

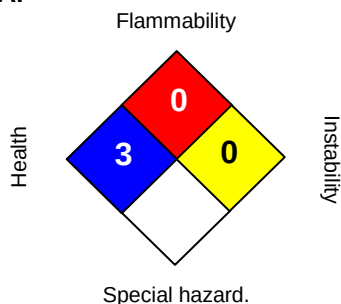
### Section: 16. OTHER INFORMATION

#### Product AS SOLD

## SAFETY DATA SHEET

### AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

#### NFPA:



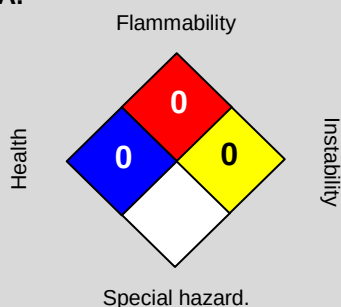
#### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

#### Product AT USE DILUTION

#### NFPA:



#### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 0 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

Issuing date : 02.12.2019  
Version : 1.3  
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.